CURRICULUM VITAE JunHo Song

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RESEARCH INTERESTS AND TOOLS

Ice-phobic coatings; polymers; metamaterials; molecular dynamics; finite element modeling; DFT; machine-learned interatomic potentials; AI for materials science.
Tools: LAMMPS, ANSYS APDL, Abaqus, MLIPs (MTP, CHGNet, SevenNet), Python, graph neural networks

EDUCATION

Soongsil University
M.S. in Mechanical Engineering Mar 2024 – Aug 2025

- Advisor: Prof. Kyoungmin Min (Yonsei University) & Prof. Heonjun Yoon (Soongsil University)
- Thesis: Analyzing the Mechanical Behavior of Zeolite-Inspired Porous Metamaterials

Soongsil University
B.S. in Mechanical Engineering Mar 2018 – Feb 2024

- Advisor: Prof. Kyoungmin Min (Soongsil University)

PUBLICATIONS

Published Articles (*indicates equal contribution)
First Authors

- **J6:** S. Kang*, **J. Song***, J. Jeong, T. Park, J. Won, J. Han, K. Min (2025). Evaluating Machine Learning Interatomic Potentials for Accurate and Scalable Modeling of Organometallic Precursors. *ACS Applied Materials & Interfaces*, 17(41), 57226–57239. https://doi.org/10.1021/acsami.5c09107
- **J5:** D. Lee*, **J. Song***, K. Min, N. Kim (2025). Investigating the Compressive Behavior of Zeolite-based Porous Mechanical Metamaterials. *Materials & Design*, 249, 113541. https://doi.org/10.1016/j.matdes.2024.113541
- **J4: J. Song***, D. Lee*, N. Kim, K. Min (2024). Molecular Dynamics and Machine Learning Insights into the Mechanical Behavior of Zeolites under Large Deformation. *Materials Today Communications*, 41, 110922. https://doi.org/10.1016/j.mtcomm.2024.110922
- **J3:** J. Sun*, **J. Song***, J. Kim*, S. Kang, E. Park, S. Seo, K. Min (2024). Enhancing protein aggregation prediction: a unified analysis leveraging graph convolutional networks and active learning. *RSC Advances*, 14, 31439–31450. https://doi.org/10.1039/D4RA06285J
- **J2:** T. Park*, **J. Song***, J. Jeong, S. Kang, J. Kim, J. Won, J. Han, K. Min (2024). Interpretable Machine Learning Boosting the Discovery of Targeted Organometallic Compounds with Optimal Bandgap. *Materials Today Advances*, 23, 100520. https://doi.org/10.1016/j.mtadv.2024.100520
- **J1: J. Song***, J. Lee*, N. Kim, K. Min (2024). Artificial Intelligence in the Design of Innovative Metamaterials: A Comprehensive Review. *International Journal of Precision Engineering and Manufacturing*, 25, 225–244. https://doi.org/10.1007/s12541-023-00857-w

Co-Author Articles (*indicates equal contribution)

- **J6:** A. Jeong, S. Kang, T. Ko, **J. Song**, Y. Jung, W. Shin, J. Sun, K. Min (2026). Machine Learning-Guided Screening of Phase-Stable High-Entropy Na Cathodes to Enable EVs and Low-Cost Charging Storage Systems. *Journal of Materials Chemistry A*, 14, 16693–16712. https://doi.org/10.1039/d6ta00306k
- **J5:** J. Lee, **J. Song**, J. Kim, A. Lee, S. Oh, B. Shin, K. Min, W.-H. Yeo (2026). Capillary-Driven, Superhydrophilic Microfluidic Retainer for Continuous Salivary Glucose Monitoring. *Lab on a Chip*, 26, 273–285. https://doi.org/10.1039/d5lc00934k
- **J4:** J. Hwang, T. Park, J. Jeong, S. Kang, **J. Song**, J. Kim, A. Jeong, W. Shin, J. Han, J. Won, K.

Min (2025). Data-Driven Curation of Organometallic Precursor Boiling Points via Natural Language Processing. *Materials & Design*, 260, 115253. https://doi.org/10.1016/j.matdes.2025.115253

- **J3:** J. Sun, **J. Song**, J. Kim, J. Jeong, E. Kim, J. Kim, T. Park, K. Min (2025). Cation-Disordered High-Entropy Garnet Structures as Solid-State Electrolytes for All-Solid-State Batteries: Machine Learning-Driven Discovery. *ACS Applied Materials & Interfaces*, 17(35), 49472–49484. https://doi.org/10.1021/acsami.5c09935
- **J2:** J. Jeong*, T. Park*, **J. Song**, S. Kang, J. Won, J. Han, K. Min (2024). Integrating Data Mining and Natural Language Processing to Construct a Melting Point Database for Organometallic Compounds. *Journal of Chemical Information and Modeling*, 64(19), 7432–7446. https://doi.org/10.1021/acs.jcim.4c01254
- **J1:** J. Kim, M. Hong, **J. Song**, N. Kim, K. Min (2023). Correlating Atomistic Characteristics of Zeolites to their 3D-Printed Macro Structural Properties for Prediction of Mechanical Response. *Materials & Design*, 233, 112189. https://doi.org/10.1016/j.matdes.2023.112189

PRESENTATIONS

Contributed Presentations

- **CP5**: **J. Song** and K. Min, "Development of an Interpretable Interatomic Potential for Predicting Fundamental Properties of Graphene," KSME Annual Meeting, Nov 2026 (accepted)
- **CP4**: **J. Song**, K. Nam, S.-H. Park, and K. Min, "Molecular Dynamics Investigation of Ice Adhesion–Slippage Transition on PDMS-Based Icephobic Surfaces," KSME CAE & Applied Mechanics Conf., Yeosu, May 2026 (poster)
- **CP3**: **J. Song**, K. Nam, S.-H. Park, and K. Min, "Exploring the Interplay Between Ice Adhesion and Slippage on Icephobic Surfaces Using Molecular Dynamics," KSME Annual Meeting, Dec 2025 (oral)
- **CP2**: T. Park, **J. Song** (presenter), J. Jung, J. Kim, S. Kang, J. Won, C. Baik, J. Han, and K. Min, "Harnessing Feature-Driven Insights for Accelerated Search for Optimal Organometallic Compounds," KSME CAE & Applied Mechanics Conf., Jeju, May 2024 (oral)
- **CP1**: **J. Song**, J. Lee, N. Kim, and K. Min, "Prediction of Atomic to Macro Level Mechanical Response Using Concatenation Networks and Active Learning," KSME Annual Meeting, Nov 2023 (oral)

PATENT

- Method and Apparatus for Predicting Protein Aggregation of Target Proteins (South Korea)

HONORS AND SCHOLARSHIPS

Graduate Entrance Merit Scholarship (3 semesters), Soongsil University — Mar 2024 – Aug 2025
Seoul Hope Scholarship, Seoul Scholarship Foundation — 2021 – 2023

RESEARCH EXPERIENCE

Researcher — Sep 2025 – Present
CS-AI Lab (PI: Prof. Kyoungmin Min), School of Mechanical Engineering, Yonsei University

- MD of ice adhesion and slippage on PDMS-based icephobic surfaces (in progress; CP3, CP4).

Graduate Research Assistant — Mar 2024 – Aug 2025
Soongsil University, Department of Mechanical Engineering

- MD, FEA, and machine learning for the mechanics of zeolites and zeolite metamaterials (J4, J5).
- MLIP benchmarks for ALD precursors (J6); ML for band gaps and protein aggregation (J2, J3).

Teaching Assistant, Solid Mechanics Laboratory — Mar 2024 – Aug 2024
Soongsil University, School of Mechanical Engineering

- Undergraduate lab: Izod impact, Rockwell hardness, beam buckling, and ultrasonic NDT tests.

Administrative Assistant (part-time) — Mar 2024 – Feb 2025
Soongsil University, School of Mechanical Engineering; College of Engineering Academic Affairs Office

Mandatory Military Service, Republic of Korea Army — Jun 2019 – Jan 2021